Tyler **Kuipers**

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About Me

I've been writing software for about 10 years, with the last three building and owning platform services, APIs, and cloud infrastructure. I work primarily in Java and JVM (via Gosu, a JVM language used by Guidewire), Python, and TypeScript. AWS and Kubernetes are where most of my production work lives: Lambda, EKS, S3, SQS, IAM, Aurora/RDS across multiple AWS accounts. At AMA I own 4 REST APIs serving 5 internal teams and external partners, built a self-service developer platform that took environment provisioning from 2-3 weeks to 2 hours (60x), and led a rewrite of a legacy system into containerized microservices on EKS. I write the RFCs that align teams on architecture and treat observability and production reliability as non-negotiable.

Work Experience

Alberta Motor Association
Edmonton, Alberta
Software Developer IV
Jan 2023 - Present

- Architected the infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, 99.9% uptime since launch, 2.8M insured trips and 76,637 invoices processed. Won the 2024 Guidewire Innovation Award (1 of 3 worldwide). (Python, TypeScript, AWS Lambda, Terraform)
- Own 4 REST APIs serving 5 internal teams and external partners. Designed the contracts and versioning, approve new endpoint additions, and maintain OpenAPI design standards in Confluence. (Python, TypeScript, REST, OpenAPI)
- Set technical standards for a 12-person team through code reviews, RFC authorship, and PR checklists / API design guides. Authored an RFC moving the team to trunk-based development with feature flags. Worked in cross-functional pods alongside PM, design, and QA. Member of the Architecture Steering Committee, and mentor newer developers through legacy codebases and TDD. (Gosu/Java)
- Built a self-service developer platform solo that took environment provisioning from 2-3 weeks down to 2 hours (60x). Self-service create/destroy, database snapshots, and pull-from-prod, adopted by 15 people across 3 teams and used to spin up 120+ environments. It made production debugging routine: a cancellation bug nobody could reproduce got root-caused in days using snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Led a 2-3 engineer rewrite of a brittle legacy C# document generation system into containerized ASP.NET Core microservices (.NET 8). 4-5 services on a managed Kubernetes cluster, deployed with Helm. Wired up Datadog metrics for failure rate, throughput, generation time, and storage time. (C#, ASP.NET Core, Kubernetes, Docker, Helm, Datadog)
- Built an event-driven pipeline that ingests an external partner's policy data: S3 ingestion, a Kubernetes CronJob, a translation service, and transactional calls to a composite API, with OAuth 2.0 auth and Datadog monitoring. (Python, Kubernetes, AWS S3, OAuth 2.0)

Arterys Inc. (Acquired by Tempus Labs)
Calgary, Alberta (Remote)
Software Developer
Aug 2021 - Aug 2022

- Built and maintained backend services that ran medical AI algorithms on clinical imaging data and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

Alberta Motor Association
Edmonton, Alberta
Software Developer | Software Team Lead
Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team (15 developers / all Guidewire projects) and still in use 4+ years later. (Gosu/Java)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

Pleasant Solutions
Edmonton, Alberta
Software Developer
Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

Lockheed Martin Canada
Halifax, Nova Scotia
Associate Software Engineer
Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

The King's University — *Edmonton, Alberta*
SOFTWARE DEVELOPER/RESEARCH ASSISTANT — *May 2014 - Sep 2016*

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

The Open University — *Milton Keynes, England (Remote)*
MSc. COMPUTING (SOFTWARE ENGINEERING) — *May 2024 - Ongoing*
Currently pursuing Master's in Computing at The Open University.

The King's University — *Edmonton, Alberta*
B.S. IN COMPUTER SCIENCE — *Sep 2012 - Dec 2016*

Skills

Backend Services & Ownership	I've spent the last three years as the technical owner of a platform handling over $300M in annual transactions, building APIs, infrastructure automation, and distributed systems. I own things end to end, from the design doc through the part where they have to stay up. I built a self-service developer platform from scratch that replaced a 2-3 week provisioning process with a 2-hour one, and I architected the backend for a pay-as-you-go product that has maintained 99.9% uptime since launch.
Reliability & Observability	I instrument what I ship with Datadog metrics for failure rate, throughput, and latency, and use them to set clear bars for what healthy looks like. I like chasing hard production failures to root cause: a cancellation bug nobody could reproduce got fixed in days once I could replay real production data against it. To me reliability is part of the feature, not a follow-up.
Technical Leadership & Design Docs	I write the design docs and RFCs that align a team on the hard calls, including an RFC to move us to trunk-based development with feature flags so we could decouple deploys from releases. I set standards through code reviews, PR checklists, and API design guides, sit on AMA's Architecture Steering Committee, and mentor developers through legacy codebases and TDD.
Software Craftsmanship	TDD has been core to how I work since 2018. I designed a test framework that reduced defects by 30% on subsequent projects and got adopted by 15 developers on the insurance engineering team; it's still in use four years later. I care about unit test coverage, continuous integration, and writing code that's easy to change. When a bug gets through, I want to understand why, fix it properly, and make sure it can't happen again.

Technologies

Backend Languages	Java / Gosu (JVM), TypeScript, Node.js, Python, C# / .NET (ASP.NET Core, .NET 8), Bash
Architecture & APIs	RESTful APIs, OpenAPI, Microservices, Event-Driven Architecture, API Versioning
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM, ECR), Terraform, Kubernetes, Docker, Azure DevOps, Azure Artifacts
Databases	PostgreSQL, SQL, Aurora/RDS, Data Modelling, Redis
AI-Assisted Development	Claude Code, Cursor, GitHub Copilot, Agentic Workflows
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, SLOs, Alerting
CI/CD & Automation	CI/CD, GitHub Actions, Jenkins, GitOps, Feature Flags, Infrastructure as Code
Testing	NUnit, Moq, JUnit, TDD, Test Framework Design